

(a) Use mathematical induction on the length of a formula to show that the number of occurrences of the symbol $\wedge$ in a formula ϕ is less than or equal to the number of left brackets (in the formula.

Note: The length of ϕ is defined as the number of occurrences of connectives in ϕ .

Proof. Base Case: Suppose that the length of ϕ is 0. Then ϕ is just a propositional variable and has no occurrence of $\wedge$ or left brackets (.

Inductive Hypothesis: Suppose that for every formula of length $\leq n$, the number of occurrences of the symbol $\wedge$ is less than or equal to the number of left brackets (in the formula.

Inductive Step: Let ϕ have length $n+1$. Since ϕ has at least one connective, it cannot be a propositional variable, so by clause (ii) of the definition of a formula ϕ takes on one of these forms: $(\theta \wedge \psi)$, $(\theta \vee \psi)$, $(\theta \rightarrow \psi)$, $(\theta \leftrightarrow \psi)$, $\neg\theta$.

ϕ is of the form $(\theta \wedge \psi)$: Then θ has length $\leq n$ since $\wedge$ accounts for one of the connectives in ϕ . We know through the inductive hypothesis that the number of occurrences of the symbol $\wedge$ is less than or equal to the number of left brackets. Also ψ has length $\leq n$ since $\wedge$ accounts for one of the connectives in ϕ , so the same holds. So let the formula θ have a occurrences of $\wedge$ and b occurrences of the left bracket with $a \leq b$. Similarly, let the formula ψ have a' occurrences of $\wedge$ and b' occurrences of the left bracket with $a' \leq b'$. So then the total number of occurrences of $\wedge$ in ϕ is $a + a' + 1$ and the total number of occurrences of the left bracket in ϕ is $b + b' + 1$. And so we have: $a + a' + 1 \leq b + b' + 1$ as desired.

ϕ is of the form $\neg\theta$: As θ has length $\leq n$ since $\neg$ accounts for one of the connectives in ϕ , let the formula θ have a occurrences of $\wedge$ and b occurrences of the left bracket with $a \leq b$ by the inductive hypothesis. Since $\neg\theta$ does not add any $\wedge$ or (to the string, the number of occurrences of $\wedge$ and left brackets stays as $a \leq b$.

ϕ is of the form $(\theta \vee \psi)$: Then as θ has length $\leq n$ since $\vee$ accounts for one of the connectives in ϕ , we know through the inductive hypothesis that the number of occurrences of the symbol $\wedge$ is less than or equal to the number of left brackets. Also ψ has length $\leq n$ since $\vee$ accounts for one of the connectives in ϕ so the same holds. So let the formula θ have a occurrences of $\wedge$ and b occurrences of the left bracket with $a \leq b$. Similarly, let the formula ψ have a' occurrences of $\wedge$ and b' occurrences of the left bracket with $a' \leq b'$. So then the total number of occurrences of $\wedge$ in ϕ is $a + a'$ and the total number of occurrences of the left bracket in ϕ is $b + b' + 1$. And so we have: $a + a' \leq b + b' + 1$ as desired.

For the cases where ϕ is of the form $(\theta \rightarrow \psi)$, $(\theta \leftrightarrow \psi)$ an identical argument applies as was made in ϕ is of the form $(\theta \vee \psi)$.

Therefore through the principle of mathematical induction we've shown that the number of occurrences of the symbol $\wedge$ in a formula ϕ is less than or equal to the number of left brackets (in the formula. $\square$

(b) Does the result of part (a) hold for the symbol $\neg$? Justify your answer.

Consider the formula $\neg\phi$. Suppose ϕ is just a propositional variable. Then there is one occurrence of $\neg$ and there are zero occurrences of the left bracket (. Since $1 > 0$ the result of part (a) does not hold for the symbol $\neg$.